



(Optional) Hands-on Lab: Advanced charts in Looker Studio

Estimated time needed: 30 minutes

In this lab session, you'll be instructed on utilizing third-party tools for advanced visualization techniques. Following that, you'll delve into creating a bubble chart and word cloud specifically within Looker Studio.

Software used in this lab

Like the videos in the course, for the hands-on labs, we will be using Google's **Looker Studio** as this is available at no charge.

Data set used in this lab

The data set used in this lab comes from IBM Cognos Analytics. This data set is published by IBM. You can download the data set file directly from here: [CustomerLoyaltyProgram.csv](#).

Objectives

After completing this lab, you will be able to:

- Utilize community visualizations.
- Create a word cloud.
- Create a scattered bubble chart.

Note: Sign in to Google Looker Studio and Make sure you have uploaded the dataset as shown in the previous labs. Properly uploading the dataset is essential for the successful completion of this lab exercise.

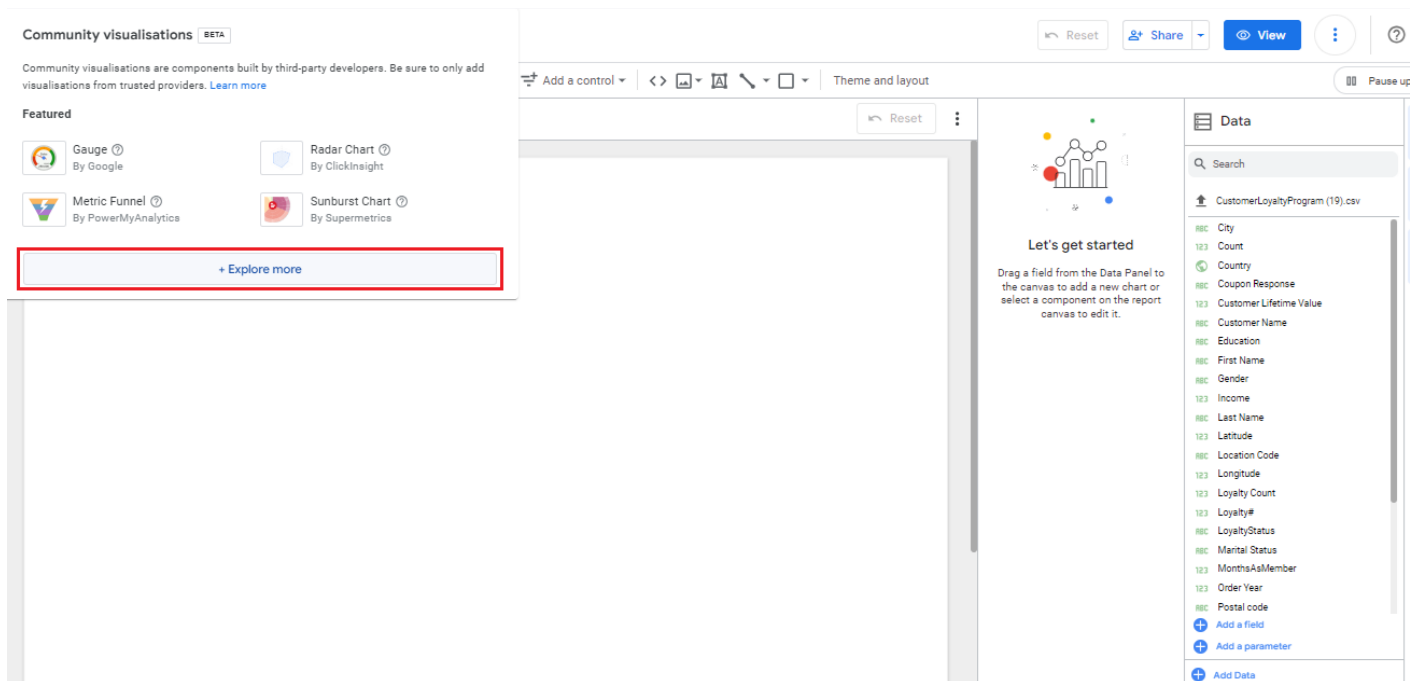
Exercise 1: Create a word cloud using community visualization tools

In this exercise, you will discover how to utilize community visualization tools to create various graphs that are not directly available in Looker Studio.

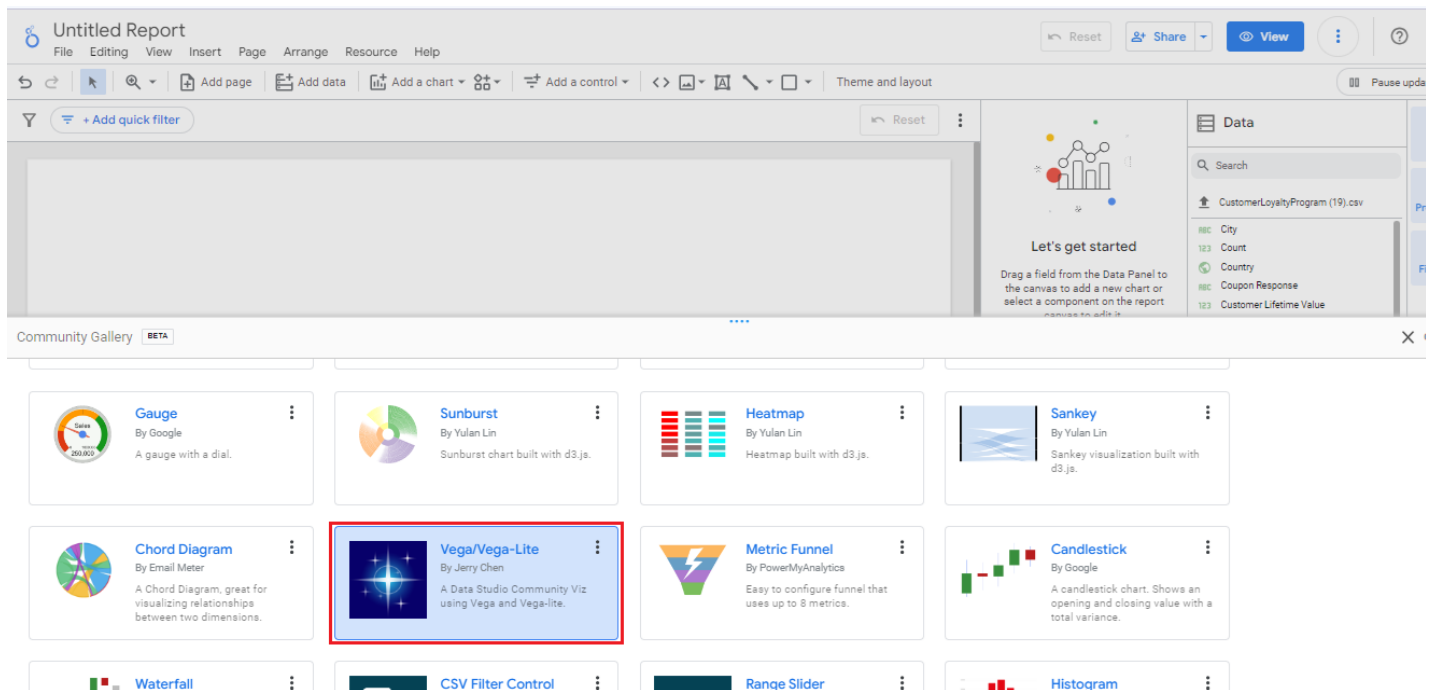
1. Start with the blank page and click on the **Community Visualization and Components**.

The screenshot shows the Google Looker Studio interface. At the top, the title is "Untitled Report". Below the title bar is a menu with "File", "Editing", "View", "Insert", "Page", "Arrange", "Resource", and "Help". A toolbar contains icons for undo, redo, select, zoom, and various chart types. The "Add a chart" button is highlighted with a red box. Below the toolbar, a "Community visualisations and components" panel is open, showing a "Let's get started" message and a list of data fields from the "CustomerLoyaltyProgram (19).csv" dataset. The fields include City, Count, Country, Coupon Response, Customer Lifetime Value, Customer Name, Education, First Name, Gender, Income, Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, and buttons to "Add a field", "Add a parameter", and "Add Data".

2. After selecting **Community Visualization and Components**, a popup will appear. Next, click **Explore More**.



3. You will be redirected to the Community Gallery. Now, scroll down and search for the **Vega/Vega-Lite** option. Select it, and grant access by clicking the Allow button.



4. After clicking on the **Vega/Vega-Lite**, a blank graph will appear on the canvas with your cursor. Click on the canvas to finalize and display the blank graph.

Untitled Report

File Editing View Insert Page Arrange Resource Help

Reset Share View

Add page Add data Add a chart Add a control Theme and layout

+ Add quick filter

Click to add or drag and draw

Let's get started

Drag a field from the Data Panel to the canvas to add a new chart or select a component on the report canvas to edit it.

Data

Search

CustomerLoyaltyProgram (19).csv

- City
- Count
- Country
- Coupon Response
- Customer Lifetime Value
- Customer Name
- Education
- First Name
- Gender
- Income
- Last Name
- Latitude
- Longitude
- Loyalty Count
- Loyalty#
- LoyaltyStatus
- Marital Status
- MonthsAsMember
- Order Year
- Postal code
- Add a field
- Add a parameter
- Add Data

5. By default, it will appear as shown in the screenshot below. We will proceed to adjust the dimensions and metrics.

Untitled Report

File Editing View Insert Page Arrange Resource Help

Reset Share View

Add page Add data Add a chart Add a control Theme and layout

+ Add quick filter

Community visualisation Properties

SET-UP STYLE

Data source

CustomerLoyaltyProgram (1...

BLEND DATA

Date Range Dimension

Add dimension

Fields

dimensions

First Name

Add dimension

metrics

Record Count

Add metric

Sort

Record Count

Descending

Ascending

Filter

Community Visualisation Filter

Data

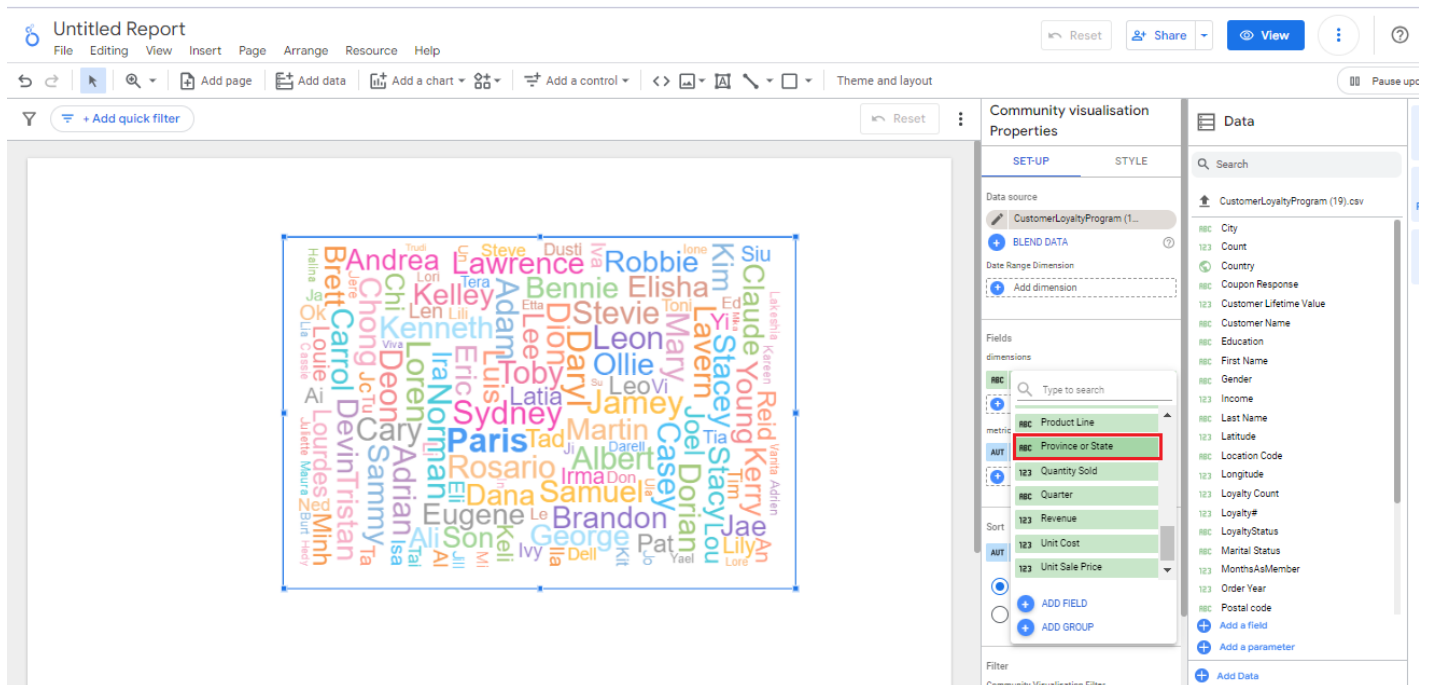
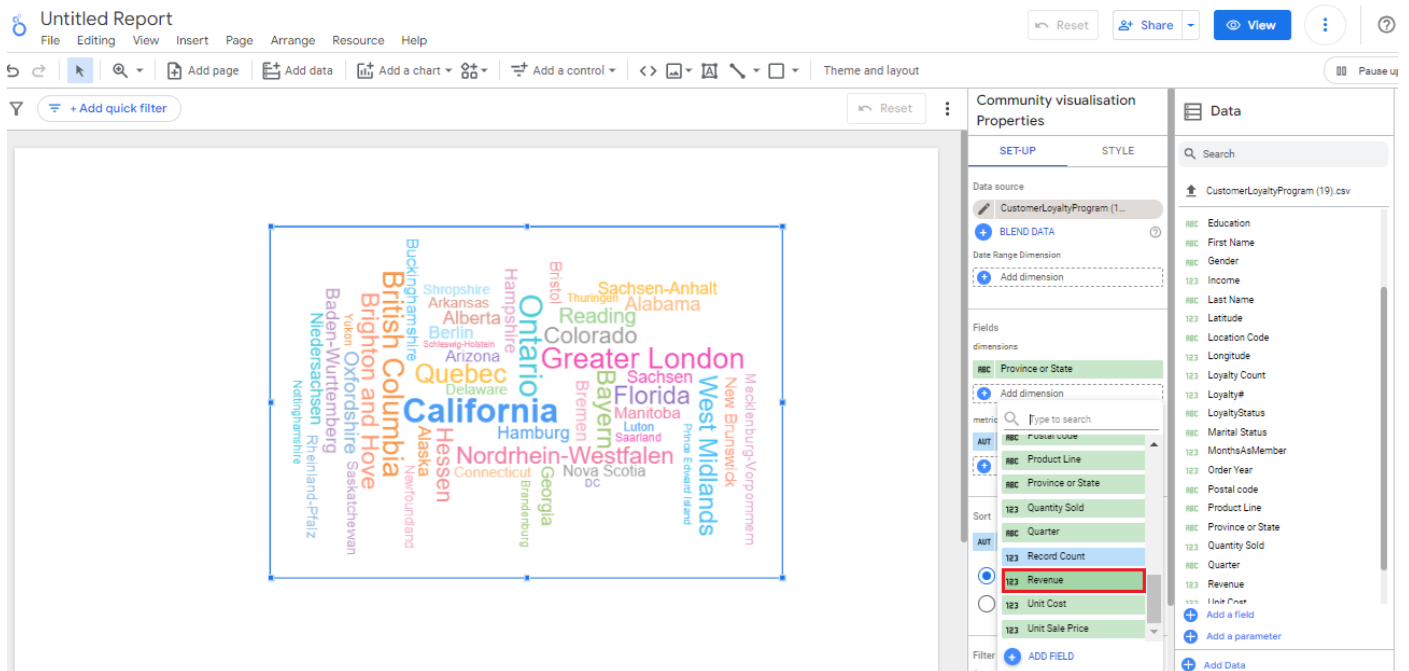
Search

CustomerLoyaltyProgram (19).csv

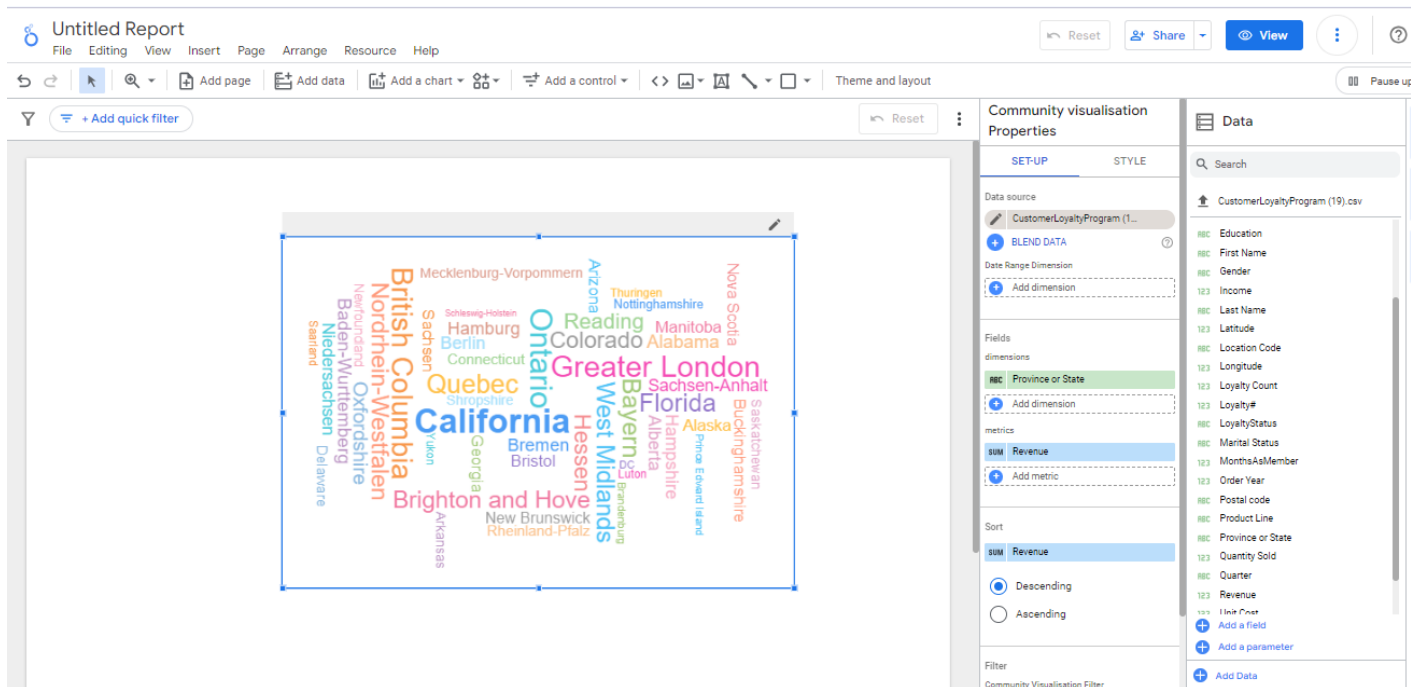
- City
- Count
- Country
- Coupon Response
- Customer Lifetime Value
- Customer Name
- Education
- First Name
- Gender
- Income
- Last Name
- Latitude
- Longitude
- Loyalty Count
- Loyalty#
- LoyaltyStatus
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- Add a field
- Add a parameter
- Add Data

6. Change the dimension from First Name to Province or State and metrics from Record count to Revenue.

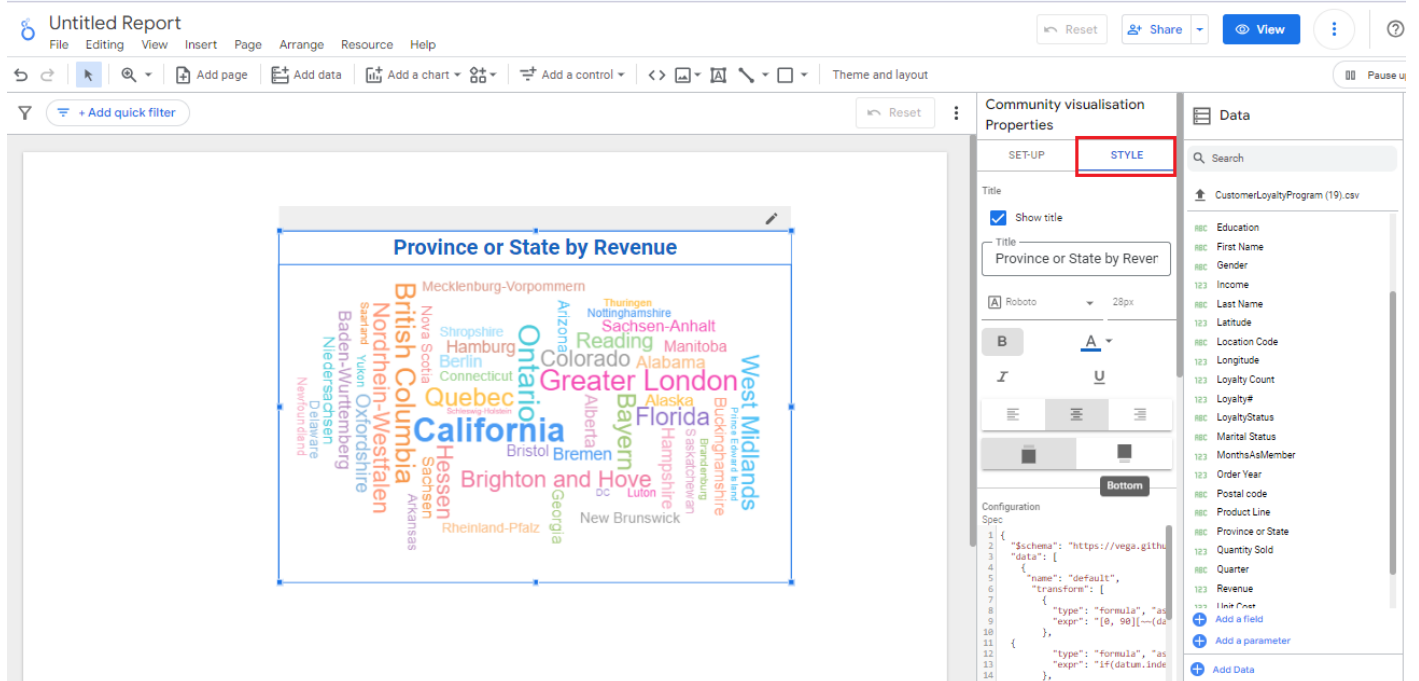
Dimension:

**Metrics:**

7. Now, this final graph represents a word cloud of **Province or State by Revenue**.



8. In the Style tab, set the title of the graph as **Province or State by Revenue** and align it in the center.



A above word cloud of **Province or State by Revenue** visualizes the relationship between different provinces or states and their corresponding revenue levels. In this visualization, the size of each province or state name within the cloud is proportional to its revenue. Typically, larger font sizes represent higher revenue values, while smaller font sizes represent lower revenue values. This type of visualization offers a quick and intuitive way to identify which provinces or states contribute the most to overall revenue and which ones contribute less.

Exercise 2: Create a Scatter bubble chart using multiple dimension

A scatter bubble chart with multiple dimensions is a powerful tool for visualizing data that has more than two dimensions. In a traditional scatter plot, you have two axes (X and Y) representing two variables, and you plot individual data points accordingly. However, in a scatter bubble chart with multiple dimensions, you can represent additional variables by using the size and color of the bubbles in addition to the X and Y coordinates.

Here, we'll create a visualization showcasing the relationship between **Product Line**, **LoyaltyStatus**, and **Quantity sold**.

1. To start, go to the toolbar and click **Add page**.
2. To add a new chart, click **Add a chart** and select a **scatter bubble chart**.

The screenshot shows the Power BI report editor interface. The top menu bar includes 'File', 'Editing', 'View', 'Insert', 'Page', 'Arrange', 'Resource', and 'Help'. The 'Add a chart' button is highlighted with a red box. The 'Data' pane on the right shows a list of fields from the 'CustomerLoyaltyProgram (19).csv' dataset, including City, Count, Country, Coupon Response, Customer Lifetime Value, Customer Name, Education, First Name, Gender, Income, Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, and Province or State. A 'Let's get started' message is displayed in the center of the canvas.

3. Click on the canvas where you want it to be positioned. Click on the scatter bubble chart in the canvas, and then click **Properties**.

4. Under the setup tab, we will change the dimensions and metrics to get the desired scatter bubble chart. From the data pane, drag **Product Line** to the Dimension field to replace **First Name**.

The screenshot shows the 'Chart' setup tab in the Power BI report editor. The 'Dimension' field is highlighted with a red box, and a red arrow points to the 'Product Line' field in the 'Data' pane. The 'Data' pane shows a list of fields from the 'CustomerLoyaltyProgram (19).csv' dataset, including Last Name, Latitude, Location Code, Longitude, Loyalty Count, Loyalty#, LoyaltyStatus, Marital Status, MonthsAsMember, Order Year, Postal code, Product Line, Province or State, Quantity Sold, Quarter, Revenue, Unit Cost, and Unit Price. The 'Product Line' field is highlighted with a green box.

5. Now add one more dimension for **LoyaltyStatus**.

Chart **SET-UP** **STYLE**

Dimension

- Product Line
- LoyaltyStatus

Metric X

- Record Count

Metric Y

- Loyalty#

Bubble Size Metric

- Latitude

Data

CustomerLoyaltyProgram (19).csv

- Last Name
- Latitude
- Location Code
- Longitude
- Loyalty Count
- Loyalty#
- LoyaltyStatus
- Marital Status
- MonthsAsMember
- Order Year
- Postal code
- Product Line
- Province or State
- Quantity Sold
- Quarter
- Revenue
- Unit Cost

Properties

Filter bar

6. Drag and drop the **Quantity Sold** into the **Bubble Size** metric area, replacing 'Latitude'.

Chart **SET-UP** **STYLE**

Dimension

- Product Line
- LoyaltyStatus

Metric X

- Record Count

Metric Y

- Loyalty#

Bubble Size Metric

- Quantity Sold

Data

CustomerLoyaltyProgram (19).csv

- Last Name
- Latitude
- Location Code
- Longitude
- Loyalty Count
- Loyalty#
- LoyaltyStatus
- Marital Status
- MonthsAsMember
- Order Year
- Postal code
- Product Line
- Province or State
- Quantity Sold
- Quarter
- Revenue
- Unit Cost

Properties

Filter bar

7. Now change the Metric X and Metric Y from **Record Count** and **Loyalty#** to **Quantity Sold** and **Revenue**.

Chart **SET-UP** **STYLE**

Dimension

- Product Line
- LoyaltyStatus

Metric X

- Quantity Sold

Metric Y

- Revenue

Bubble Size Metric

- Quantity Sold

Data

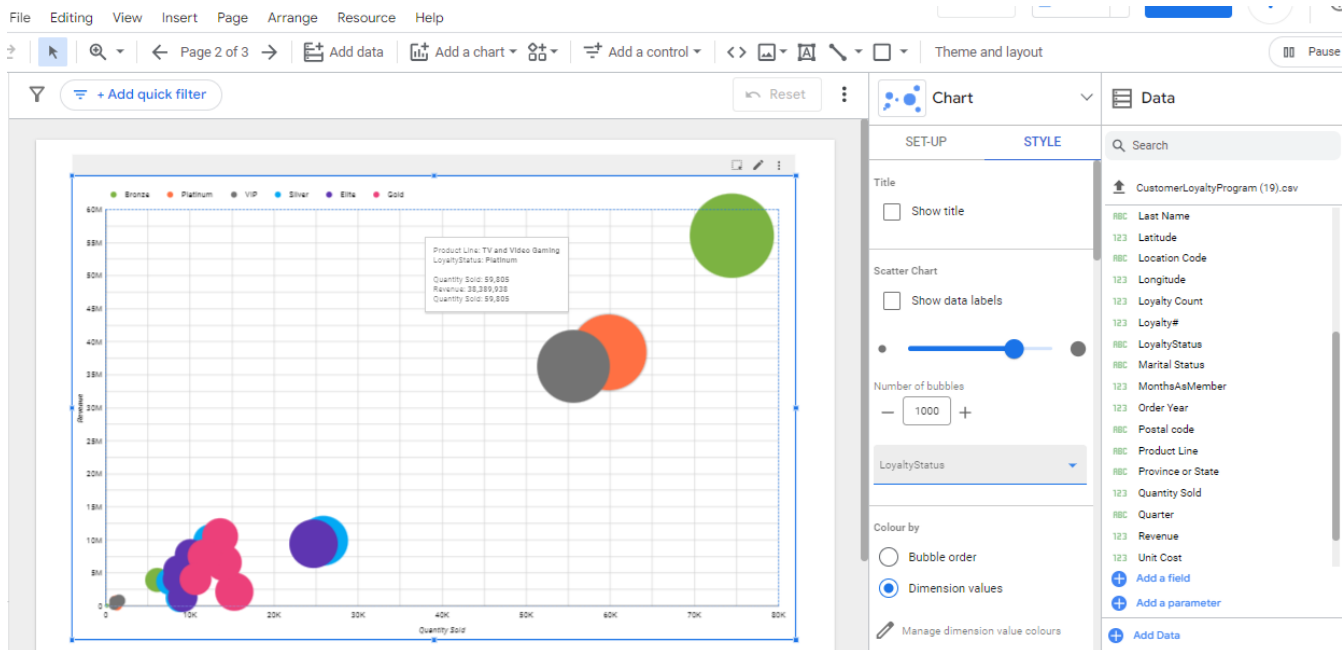
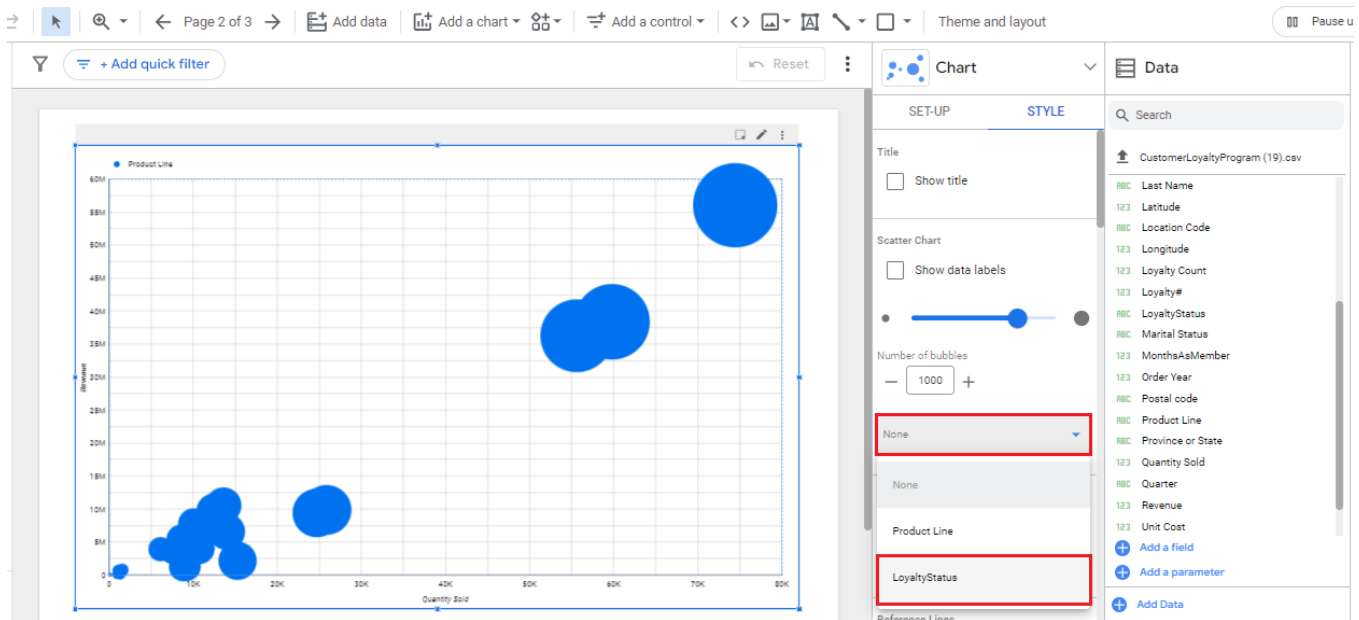
CustomerLoyaltyProgram (19).csv

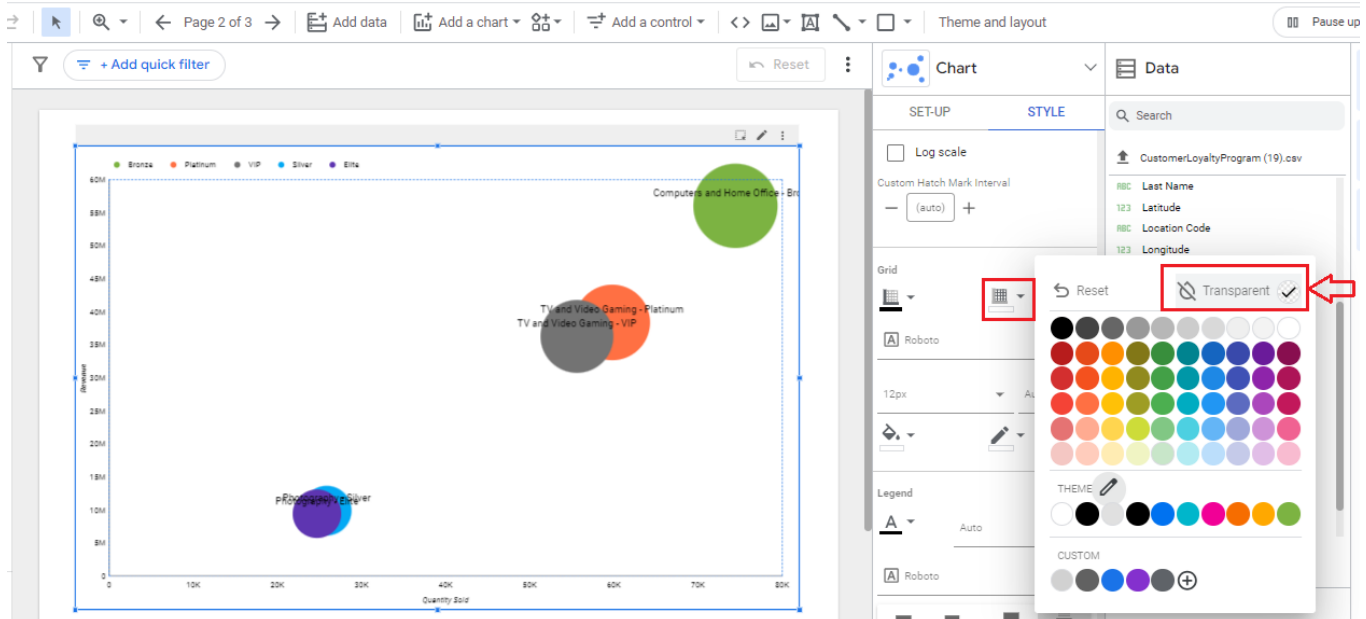
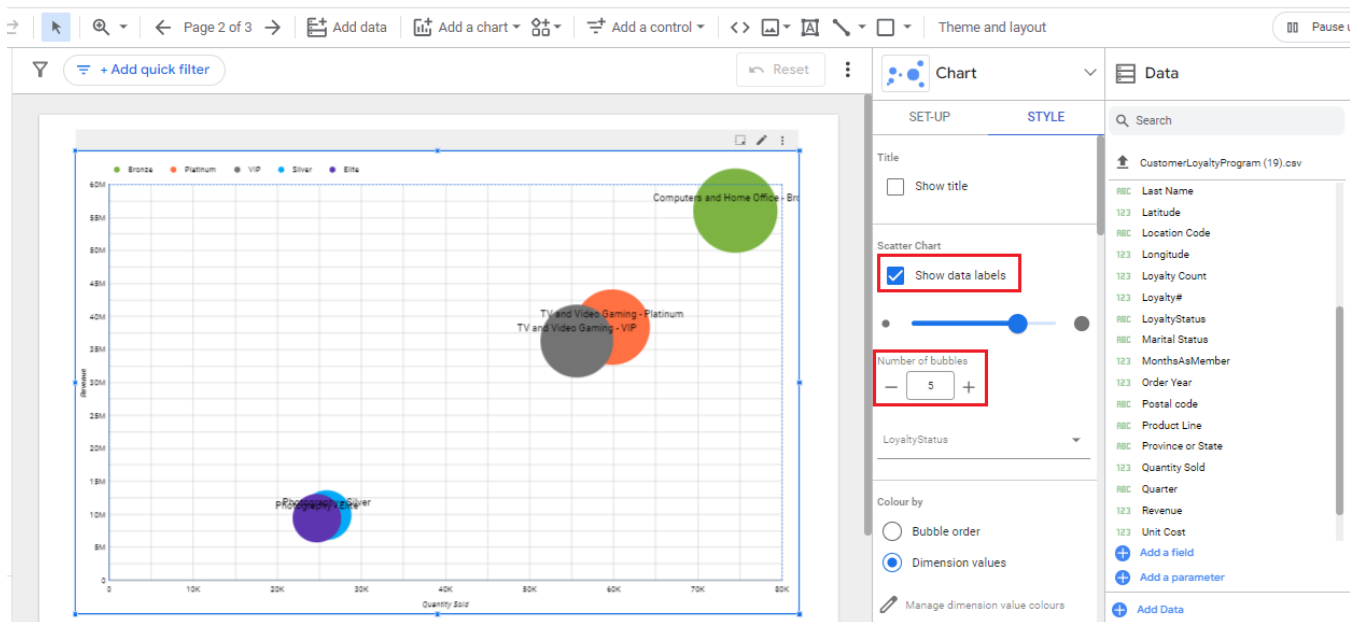
- Last Name
- Latitude
- Location Code
- Longitude
- Loyalty Count
- Loyalty#
- LoyaltyStatus
- Marital Status
- MonthsAsMember
- Order Year
- Postal code
- Product Line
- Province or State
- Quantity Sold
- Quarter
- Revenue
- Unit Cost

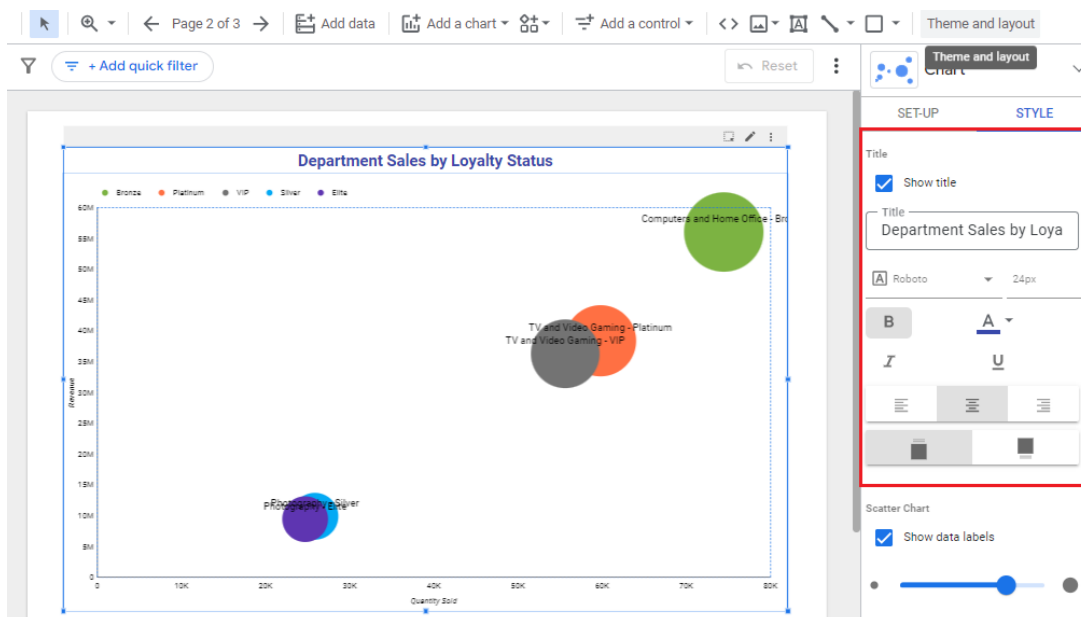
Properties

Filter bar

8. We will get the initial graph as shown below. Next, click the **STYLE** tab in the chart's Properties pane, then click on the drop down of **Bubble Color** field and select **Loyalty status**.







The scatter bubble graph, using dimensions of Product Line and Loyalty Status, and metrics of Quantity Sold (X-axis), Revenue (Y-axis), with bubble size representing Quantity Sold and bubble color indicating Loyalty Status, provides a comprehensive visualization of sales data. Each bubble corresponds to a product line, with its position on the X-axis showing the quantity sold and its position on the Y-axis showing the total revenue generated. The size of the bubble reflects the quantity sold, with larger bubbles indicating higher sales volumes. The color of the bubble represents the loyalty status of the customers, allowing for easy differentiation between various customer segments, such as Gold, Silver, or Bronze. This graph helps in understanding the performance of different product lines, the impact of customer loyalty on sales and revenue, and the relationship between the quantity sold and the revenue generated.

Congratulations! You have completed this hands-on lab and you are now ready for the next topic.

For more help, you can refer to the [Tutorial on Looker Studio by Google](#)

Author(s)

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